RESERVE: Climate-Aware Resource Routing

A Project and Evidence Report on Explainable AI, Emissions-Aware Decisions, and Cryptographic Accountability

Umar Tabbsum
IEEE ClimateChain Global Hackathon 2026
Project repository: https://github.com/u19725505-star/reserve-climatechain-2026
Live demonstration: https://reserve-climatechain.onrender.com/

***Abstract*—RESERVE (Climate-Aware Resource Routing) explores how resource-routing decisions can account for operational feasibility, estimated transport emissions, explainable prioritization, and the integrity of supporting records. Developed as a working web prototype for the IEEE ClimateChain Global Hackathon 2026, the project addresses a practical problem in sustainable resource management: selecting a feasible destination for a shipment requires more than choosing the closest facility or minimizing a single numerical score.**

The application combines eligibility checks, transport-related emissions estimates, a transparent weighted decision model, baseline comparison, and a local cryptographic audit-log mechanism. A separate research audit examines the reproducibility of the synthetic demonstration, the consistency of its baseline, and the applicability of a candidate emissions factor derived from the U.S. Environmental Protection Agency's published truck factors.

The evidence supports a specific, limited conclusion. RESERVE demonstrates a coherent climate-aware decision-support workflow, but it has not established measured real-world emissions reductions. The current demonstration uses fictional processor profiles and an illustrative, unsourced fallback factor. An EPA-derived candidate was calculated for sensitivity analysis but was not adopted because transport mode, freight allocation, and emissions-boundary applicability remain unresolved. The project is therefore best understood as a working prototype accompanied by an explicit evidence-validation process, rather than a validated operational climate-impact system.

***Index Terms*—Climate action, sustainable supply chains, resource routing, explainable decision support, emissions estimation, cryptographic audit trails, blockchain, evidence validation.**

I. Introduction

Climate action depends not only on developing new technologies, but also on improving the everyday decisions through which materials, energy, transport, and infrastructure are used. In recycling and resource recovery, a routing decision can influence whether a shipment reaches a suitable processor, how far it travels, and what transport emissions may be associated with that movement.

These decisions involve several constraints at once. A facility may be geographically close but lack sufficient capacity. Another may accept the required material but fail a quality requirement or be unable to meet the delivery window. A routing system that considers distance without checking feasibility may recommend an option that cannot actually be used. Conversely, a system that checks feasibility but ignores transport-related emissions misses an opportunity to make the environmental consequences of a decision more visible.

RESERVE was developed to explore this problem through a working decision-support prototype. Its purpose is to bring feasibility screening, emissions estimation, explainable ranking, and record traceability into one workflow. The project is not intended to suggest that software alone can solve the climate challenge. Rather, it examines how a transparent computational process could help people make more informed resource-routing decisions.

The central proposition is straightforward: a climate-aware recommendation should be feasible, explainable, and supported by evidence. An emissions estimate should not be presented as a measured outcome, and a cryptographic record should not be treated as proof that the underlying information is true. These distinctions shape both the implementation and the research audit described in this report.

II. The IEEE ClimateChain Challenge

The IEEE ClimateChain Global Hackathon 2026 is presented by its organizers under the theme "AI and BLOCKCHAIN: Innovate Today for a Sustainable Tomorrow." The challenge invites participants to develop prototypes that use AI, blockchain, and data-driven approaches to address real-world climate problems. Its stated areas include sustainable supply chains, carbon-market and emissions transparency, renewable energy, and climate data and environmental monitoring [1].

The sustainable supply-chain track is especially relevant to RESERVE. The challenge description identifies traceability, transparency, recycling systems, and circular-economy applications as potential areas for innovation. The climate-data track is also relevant because environmental decisions are only as credible as the data and assumptions supporting them.

The challenge is not simply to place AI and blockchain into an application. The technologies should have a meaningful role in addressing the climate problem. For RESERVE, that role is to help compare feasible resource-routing options, expose the assumptions behind emissions estimates, and demonstrate how decision records could be made more traceable.

A. How RESERVE relates to the challenge

Table I connects the hackathon's stated aims with the work completed and the evidence currently available. This distinction

matters: implementation of a prototype feature does not by itself establish real-world climate impact.

TABLE I
RESERVE'S ALIGNMENT WITH THE CLIMATECHAIN CHALLENGE

Challenge area	RESERVE contribution and status
Sustainable supply chains	Screens candidate processors against shipment requirements and ranks feasible options. Demonstrated with synthetic profiles.
AI-assisted decisions	Uses a transparent weighted multi-criteria model to explain the relative ranking of eligible options. It is not trained ML.
Emissions transparency	Calculates transport-related estimates from shipment mass, distance, and a supplied factor. The current fallback is illustrative and unsourced.
Evidence accountability	Includes a local SHA-256 hash-chain mechanism and a Solidity registry prototype. Public-chain deployment and live transactions have not been established.
Climate credibility	A separate research audit checks scenario reproducibility, baseline logic, factor sensitivity, and unsupported claims. Operational climate impact remains unverified.

The project is therefore most directly aligned with sustainable supply chains, with a supporting contribution to climate-data transparency. Its significance lies in connecting the decision workflow to a disciplined treatment of the evidence behind its environmental estimates.

III. RESERVE AS A WORKING PROTOTYPE

The public RESERVE repository contains the application logic, demonstration data, decision-ranking code, evaluation components, tests, and a prototype Solidity contract [2]. The live website makes the concept accessible without requiring a reader to reconstruct the project from source code alone [3].

The intended workflow begins with a shipment request. The system checks candidate processors against constraints such as accepted material, available capacity, quality, and delivery timing. Only feasible candidates should proceed to ranking. For those candidates, the application estimates transport-related emissions using shipment mass, distance, and a configured factor. A weighted model can then compare several criteria and provide an advisory recommendation.

The distinction between eligibility and ranking is fundamental. Eligibility rules answer whether a processor can satisfy the shipment requirements. Ranking addresses which of the eligible options is preferable under the selected criteria. A high score must not override a failed hard constraint.

The application is a prototype rather than a production logistics platform. Its current demonstration profiles are fictional, and its distance values are scenario inputs rather than verified route measurements. The website should therefore be understood as a demonstration of the decision process, not as evidence that live processor networks or operational shipment systems have been validated.

A. The decision workflow

The main process can be summarized as follows:

1) Define the shipment's material, quantity, quality requirement, and delivery window.
2) Exclude processors that fail a hard eligibility rule.
3) Estimate transport-related emissions for eligible options.
4) Compare eligible options using transparent weighted criteria.
5) Explain the selected option and compare it with a defensible baseline.
6) Preserve relevant decision information in an audit record.
7) Check whether the data and assumptions justify any environmental claim.

This separation makes the decision process easier to inspect. It also creates a clear place to introduce verified processor records, route data, and applicable emissions factors when such evidence becomes available.

IV. DEMONSTRATION SCENARIO AND RESULTS

The research audit reconstructed a demonstration shipment of 10 metric tonnes of recycled plastic. The shipment requires a minimum quality score of 0.80 and a delivery window spanning days 5 through 8. Four fictional processor profiles are evaluated.

TABLE II
SYNTHETIC PROCESSOR SCENARIO USED IN THE AUDIT

Processor	Capacity (t)	Quality	Distance (km)	Scenario result
A	25	0.92	25	Eligible
B	18	0.86	15	Eligible
C	6	0.95	8	Capacity insufficient
D	40	0.98	32	Material not accepted

Processor C is the closest option in this scenario, but its capacity of 6 tonnes is insufficient for a 10-tonne shipment. Processor D has adequate capacity but accepts wood rather than the required recycled plastic. Neither should be ranked as a valid destination for this shipment. Processors A and B satisfy the demonstrated eligibility constraints.

This example illustrates why distance alone is not a complete routing policy. A climate-aware system must first identify which options are operationally feasible. Otherwise, a comparison can appear environmentally attractive while relying on a destination that cannot accept the shipment.

Figure 1 illustrates the first safeguard in the workflow: the system should not allow a lower distance or a higher quality score to compensate for a failed eligibility condition. The feasibility screen is important because it keeps the subsequent emissions and score comparisons focused on options that can actually meet the scenario's requirements.

V. WHAT AI CONTRIBUTES TO RESERVE

In RESERVE, the role of AI-assisted decision support is to make a multi-criteria routing decision more structured and explainable. The current implementation uses a deterministic weighted multi-criteria decision model. It is not a trained

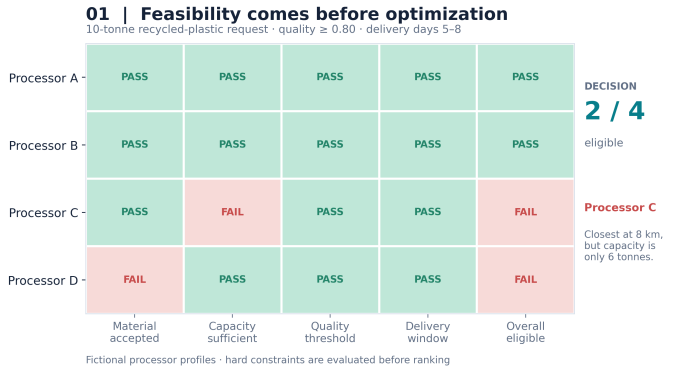


Fig. 1. Feasibility matrix for the synthetic shipment. The intended interpretation is that hard constraints determine which processors are eligible before the ranking model is applied. Processor C fails the capacity requirement, while Processor D fails the material compatibility requirement. These are demonstration records, not verified operational processor data.

machine-learning model, and it has not been calibrated against operational shipment outcomes [4].

The model considers four criteria: transport-related emissions, quality margin, capacity headroom, and delivery-window fit. The configured weights are shown in Table III.

TABLE III
WEIGHTS USED BY THE ADVISORY DECISION MODEL

Criterion	Weight
Transport-related emissions	0.45
Quality margin	0.20
Capacity headroom	0.20
Delivery-window fit	0.15
Total	1.00

The model's output is an advisory score, not an autonomous decision that has been proven optimal. The emissions criterion receives the largest weight, but the remaining weights also affect the relative ranking. Hard eligibility rules remain a prerequisite: the model should compare eligible options rather than use a high composite score to excuse an infeasible one.

For the reconstructed scenario, Processor B receives a score of approximately 80.7%, compared with 55.0% for Processor A. The result reflects the model's configured weighting and synthetic inputs. It should not be interpreted as an 80.7% probability that Processor B is the best real-world choice, nor as an 80.7% reduction in emissions.

The score decomposition in Figure 2 is valuable because it allows a user to inspect the basis of the recommendation. A transparent model does not eliminate uncertainty, but it makes assumptions and trade-offs more visible. This is particularly important in climate-related applications, where a confident-looking numerical result can otherwise obscure weak data or unsupported assumptions.

The next step for the AI component is not simply to make the model more complex. It is to collect reliable operational data, justify the weights with stakeholders or measured outcomes, evaluate the model against relevant baselines, and test whether its recommendations remain useful across different shipment conditions.

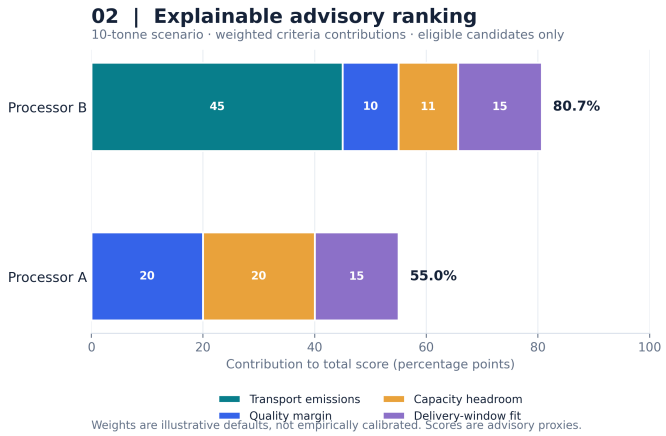


Fig. 2. Explainable multi-criteria score comparison for the demonstration. Processor B scores approximately 80.7% and Processor A 55.0%. The figure is intended to show how several weighted criteria contribute to an advisory ranking. The weights are illustrative, the profiles are synthetic, and the scores are not validated predictions of operational performance.

VI. Emissions Estimation and Fair Baseline Comparison

The current demonstration estimates transport-related emissions using a simple activity-based relationship:

$$E = m \times d \times f, \tag{1}$$

where E is estimated emissions in kilograms of CO_2 equivalent, m is shipment mass in metric tonnes, d is the entered one-way distance in kilometres, and f is the factor in kg CO_2e per tonne-kilometre.

This calculation is meaningful only when the units and scope of the factor match the shipment activity. A factor for a particular truck category, freight allocation method, or emissions boundary should not automatically be applied to every type of transport.

The current fallback factor in the application is 0.1 kg CO_2e/(tonne$-$km). The repository identifies this value as illustrative and unsourced. It is therefore appropriate for demonstrating the calculation, but not for presenting a verified environmental footprint.

A. Results using the demonstration factor

For the 10-tonne shipment, Processor A is 25 km away and Processor B is 15 km away. Under the illustrative factor of 0.1, their respective estimates are:

$$E_A = 10 \times 25 \times 0.1 = 25 \text{ kg } CO_2\text{e}, \tag{2}$$

$$E_B = 10 \times 15 \times 0.1 = 15 \text{ kg } CO_2\text{e}. \tag{3}$$

These values are calculated outputs from synthetic inputs. They are not measured emissions from actual shipments.

Figure 3 shows the immediate effect of the distance input under a common positive emissions factor. In this scenario,

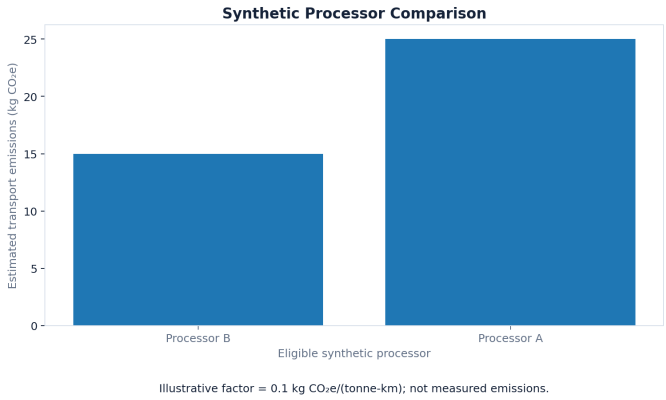


Fig. 3. Illustrative transport-emissions comparison for the synthetic 10-tonne shipment. Processor A is assigned 25 km and Processor B 15 km; the fallback factor of 0.1 produces estimates of 25 and 15 kg CO_2e, respectively. The plot illustrates the calculation and ranking, not measured emissions.

the shorter-distance eligible processor has the lower estimated transport emissions. This result is conditional on the chosen factor and the entered distances; it is not a general demonstration that the nearest processor always produces the lowest total environmental impact.

B. Why the baseline matters

An emissions-aware recommendation is not enough to demonstrate an environmental benefit. A claimed reduction requires a defensible comparison with what would otherwise have happened. The comparison must use equivalent shipment requirements and feasible alternatives.

For this scenario, the nearest eligible processor is Processor B at 15 km. The decision replay also selects Processor B. Under the same illustrative factor, the selected option and the nearest-eligible baseline both estimate 15 kg CO_2e. The modeled difference is therefore zero.

TABLE IV
BASELINE RECONCILIATION FOR THE SYNTHETIC SCENARIO

Measure	Result
Selected processor	B
Nearest eligible baseline	B
Selected estimate	15 kg CO_2e
Baseline estimate	15 kg CO_2e
Modeled difference	0 kg CO_2e
Modeled reduction	0%

Figures 4 and 5 make an important point about evaluation discipline. A system should not claim emissions savings simply because it selects an option. If the selected option is already the nearest eligible processor, the baseline comparison provides no modeled reduction. This is a meaningful result because it shows that the evaluation can report zero improvement rather than manufacture a positive climate claim.

VII. EMISSIONS-FACTOR RESEARCH AND SENSITIVITY

The separate research audit examined whether a published emissions factor could provide a more defensible alternative

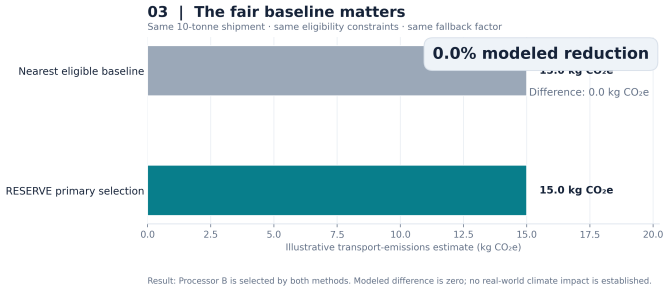


Fig. 4. Fair-baseline comparison. The selected processor is compared with the nearest processor that satisfies the same eligibility rules, rather than with the nearest facility regardless of feasibility. In this scenario, both are Processor B, so the modeled improvement is zero.

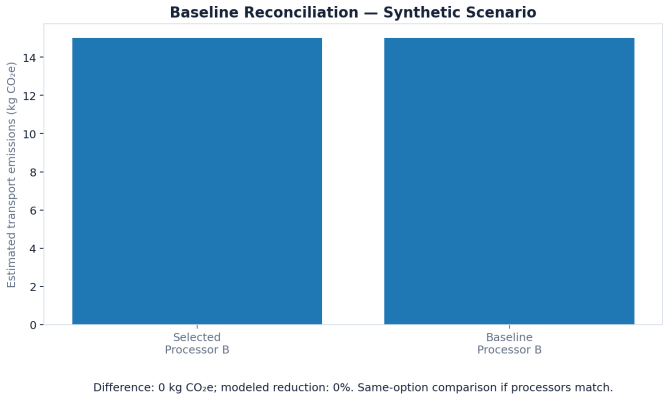


Fig. 5. Reconciliation of the replayed decision with the nearest-eligible baseline. Both estimates are 15 kg CO_2e under the illustrative factor, giving a modeled difference of zero and no demonstrated reduction in this scenario.

to the illustrative fallback. The audit used the U.S. Environmental Protection Agency's 2025 GHG Emission Factors Hub workbook, including its truck factor table, together with global warming potential values from the IPCC Fifth Assessment Report [6], [7].

The calculation combined the following values for the candidate truck factor:

- CO_2: 0.186 kg per short ton-mile.
- CH_4: 0.0016 g per short ton-mile.
- N_2O: 0.0054 g per short ton-mile.
- Global warming potentials: 1 for CO_2, 28 for CH_4, and 265 for N_2O.

After converting the gas contributions to a common mass unit and applying the stated global warming potentials, the combined value is 0.1874758 kg CO_2e per short ton-mile. Using the conversion factor applied in the audit, this corresponds to approximately 0.128410516861 kg CO_2e per metric tonne-kilometre.

The result is a candidate factor for sensitivity analysis, not a factor that has been adopted by the application. It represents a truck-combustion, or tank-to-wheel, boundary under the source assumptions; upstream fuel-cycle emissions are excluded. The transport mode and freight-allocation assumptions for the intended RESERVE use case have not been confirmed. The

calculation therefore does not, by itself, establish that this factor is appropriate for the live application's scenario.

TABLE V
FACTOR PROVENANCE AND ADOPTION DECISION

Item	Finding
Current application factor	0.1 kg CO_2e/(tonne-km); illustrative and unsourced.
Candidate source	EPA 2025 GHG Emission Factors Hub, truck factor table.
Candidate result	0.128410516861 kg CO_2e/(tonne-km).
Boundary	Truck-combustion emissions; upstream fuel-cycle emissions excluded.
Adoption status	Not adopted; applicability remains unresolved.
Outstanding checks	Transport mode, freight allocation, distance definition, factor boundary, and operational baseline.

A. What the sensitivity comparison tells us

Applying the candidate factor to the same synthetic shipment gives approximately 32.10 kg CO_2e for Processor A and 19.26 kg CO_2e for Processor B. The estimates are higher than those produced by the fallback factor, but the ranking remains unchanged because the same positive factor is applied to both distances.

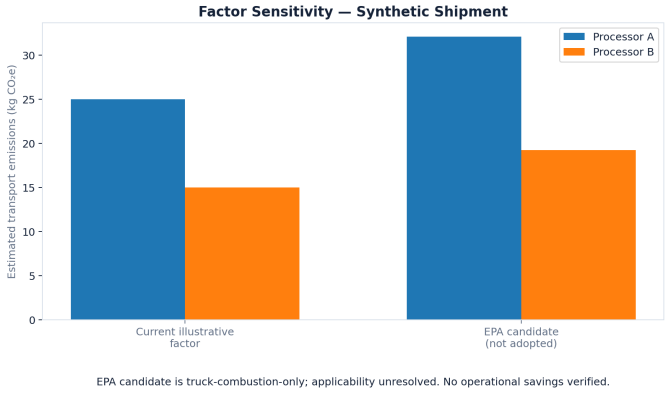


Fig. 6. Factor-sensitivity comparison using the illustrative fallback factor of 0.1 and the EPA-derived candidate of approximately 0.12841 kg CO_2e per tonne-kilometre. The candidate increases the estimated values for both processors but preserves their relative distance-based ranking. It was not adopted, and the comparison does not establish the candidate's applicability or real-world savings.

Figure 6 shows why factor provenance matters. Changing a factor can materially change the absolute emissions estimates even when it leaves the ranking unchanged. A stable ranking under a common factor is not proof that the factor is correct. It only demonstrates that the relative comparison is unchanged under that particular assumption.

The appropriate decision at this stage is to retain the illustrative behavior for the demonstration and clearly disclose its status, rather than silently replace it with a sourced value whose scope may not match the intended transport activity. Adopting a new factor should follow verification of mode, allocation method, emissions boundary, and route data, with corresponding updates to factor metadata, application behavior, tests, and demonstration outputs.

VIII. Cryptographic Accountability and the Blockchain Component

The blockchain aspect of RESERVE addresses a different question from the ranking model. The decision model helps compare eligible options; the cryptographic component explores how decision records can be linked and inspected. This distinction is important because a recommendation and its evidence trail serve different purposes.

The repository includes a local audit-log mechanism that uses SHA-256 hash chaining. In a hash chain, a record incorporates a cryptographic digest associated with the preceding record. Altering an earlier record without updating subsequent links can make the inconsistency detectable when the chain is verified. This is a useful prototype technique for demonstrating record integrity.

However, a local hash chain is not the same as a decentralized blockchain. Its assurance depends on how records are stored, who controls the environment, how verification is performed, and whether an independent copy exists. A hash can help detect certain changes to a record; it cannot prove that the original record was true, that the shipment occurred, or that the emissions factor was appropriate.

The repository also contains a Solidity registry prototype [5]. Its presence demonstrates that a smart contract component has been developed for exploration. The available evidence does not establish that the contract has been deployed to a public blockchain or that the live website is submitting transactions to it. These capabilities should not be represented as completed production functionality.

A. Why this distinction matters for climate claims

For a climate-oriented system, traceability is useful only when the information being traced is meaningful. A cryptographically linked record of an unsupported estimate remains an unsupported estimate. Likewise, placing a record on a blockchain does not independently verify the mass of a shipment, the route travelled, the processor's capacity, or the emissions associated with the transport.

The intended longer-term direction is to link decisions with supporting shipment records, source metadata, factor versions, baseline definitions, and validation results. If such evidence were later registered on a suitable public chain, independent parties could potentially inspect the registration and verify that the referenced record has not changed. That is a design direction, not a claim that the complete system currently operates in this manner.

The present contribution is therefore a prototype exploration of cryptographic accountability. Its next stage would require an explicit threat model, persistent record storage, independent verification procedures, appropriate key management, and a documented deployment and transaction workflow.

IX. The Research Audit: What Was Checked

The research audit was conducted separately from the live web application. It examined source implementation, reconstructed a synthetic shipment scenario, compared the replayed decision with a nearest-eligible baseline, reviewed the factor calculation, and recorded evidence gaps and permitted claim language.

This separation matters. The audit does not mean that its outputs are automatically part of the public website, nor does it imply that the application has been changed. The purpose was to assess what the current prototype supports and what additional evidence would be needed before stronger conclusions could be made.

TABLE VI
Evidence-readiness assessment

Evidence area	Current assessment
Decision logic	Reconstructed and reproduced for the audited synthetic scenario.
Automated tests	A saved summary reports 45 tests passing. The research audit did not independently rerun that test suite.
Processor profiles	Synthetic demonstration records; real processor data not verified.
Route distances	Scenario inputs; actual operational routes not verified.
Fallback factor	Illustrative and unsourced; not suitable as a verified factor.
EPA-derived candidate	Calculation checked for sensitivity; not adopted because applicability remains unresolved.
Baseline and savings	The audited scenario shows zero modeled improvement against the nearest eligible baseline.
Measured climate impact	Not established. No verified operational emissions reduction has been demonstrated.

The distinction between a reported test result and an independently rerun test is intentional. The saved summary reports 45 passing tests, but the audit did not rerun those tests. This report therefore records that result as previously reported rather than presenting it as a new independent test execution.

The same evidence discipline applies to the factor analysis. A calculation can be internally consistent while the selected factor remains inappropriate for a particular shipment. The audit therefore separates arithmetic checks from applicability checks and operational validation.

A. What the audit supports

The current evidence supports the following statements:

- The demonstrated workflow can screen synthetic processor profiles against the shipment constraints used in the scenario.
- The decision model can produce a transparent advisory ranking for eligible candidates.
- The emissions estimates can be reproduced from the entered distances, shipment mass, and stated factor.
- The fair-baseline comparison can report zero improvement when the selected option is already the nearest eligible option.
- An EPA-derived candidate factor can be calculated and compared with the illustrative fallback without automatically adopting it.

These are meaningful prototype and audit results. They do not establish that the same ranking, factor, or emissions values apply to actual shipments. That conclusion requires operational evidence which is not currently available in the audited package.

X. Climate Claims and Responsible Reporting

Climate applications carry a particular risk: a modeled estimate can easily be presented as if it were a measured result. RESERVE's research audit treats claim wording as part of technical validation, rather than as a communications issue to be addressed after development.

Table VII summarizes the distinction between claims supported by the current evidence and claims that remain unverified.

TABLE VII
Claims supported by the present evidence

Proposed claim	Assessment and defensible wording
RESERVE screens candidate processors.	Supported for the implemented demonstration: it applies material, capacity, quality, and delivery-window checks.
RESERVE ranks eligible options using estimated emissions.	Supported for the demonstration, provided the factor and synthetic nature of the inputs are disclosed.
The EPA candidate is correct for the current scenario.	Not established. Its applicability to the intended transport mode and freight-allocation basis must be confirmed.
RESERVE has demonstrated actual emissions reductions.	Not established. The audited scenario has zero modeled improvement against its nearest-eligible baseline, and operational savings have not been verified.
RESERVE operates a public blockchain registry.	Not established by the available evidence. A Solidity prototype exists, but public deployment and live transactions are unverified.

This approach does not diminish the project. It strengthens the credibility of the claims that can be made. A prototype that clearly states what it has demonstrated and what remains unknown is easier to evaluate and improve than one that presents every calculated number as a climate outcome.

For judges, the distinction is particularly relevant to technical execution and practical usefulness. It shows that the project is not only concerned with producing a recommendation, but also with the assumptions, evidence, and evaluation methods needed to interpret that recommendation responsibly.

XI. Why RESERVE Matters

The significance of RESERVE lies in the way it brings several concerns into one decision workflow. Feasibility rules protect against invalid recommendations. The emissions estimate makes transport consequences visible. The weighted model offers an interpretable comparison among eligible options. The cryptographic audit mechanism demonstrates one

way to link records, while the separate evidence audit asks whether the inputs and resulting climate claims are defensible.

The individual methods are not presented as new scientific discoveries. The project's contribution is their integration into an accessible prototype with an explicit distinction between decision support and verified environmental impact.

This integration matters because a climate decision can fail in more than one way. A technically correct optimization may use inaccurate input data. An emissions calculation may use a factor that does not match the transport activity. A cryptographic record may preserve information without proving that it is true. A persuasive dashboard may show a positive difference against an unfair baseline. RESERVE's evidence-oriented approach brings these risks into view.

The audit's zero-reduction result is a useful illustration. It would have been easy to highlight the difference between Processor A and Processor B and describe it as a reduction. However, the audited baseline selects Processor B as the nearest eligible alternative, and the replay also selects Processor B. The defensible result is zero modeled improvement against that baseline. Reporting this outcome demonstrates that the evaluation is not designed merely to produce a positive climate narrative.

The next opportunity is to extend the same discipline to real operational records. With verified shipment masses, actual routes, documented transport modes, suitable emissions factors, and defensible counterfactuals, RESERVE could be evaluated on whether its recommendations improve decisions under real conditions. Until then, its environmental value should be described as potential and its demonstrated contribution as a reproducible prototype workflow.

XII. Assessment Against the Challenge Objectives

The official ClimateChain challenge describes judging dimensions including climate impact, innovation and creativity, technical execution, practical usefulness, and presentation and communication [1]. RESERVE can make a credible case in each area by connecting its design to the evidence available.

The strongest present case is technical and methodological: RESERVE provides a working prototype and a research audit that examines the credibility of its outputs. The climate-impact case remains prospective until actual operations and environmental outcomes are measured.

This is the appropriate distinction for a challenge that asks participants to build prototypes addressing real-world climate problems. The prototype establishes a direction and a testable workflow; the validation roadmap defines what must happen before the potential environmental benefit can be claimed as an observed result.

XIII. Limitations

Several limitations define the present scope of the project.

First, the processor profiles and distances used in the audited scenario are synthetic. They do not establish the availability, capacity, accepted materials, or route distances of actual facilities. Second, the current fallback emissions factor is illustrative and unsourced. The EPA-derived candidate has not been adopted because its transport mode, allocation basis, and boundary have not been confirmed for the intended application.

TABLE VIII
Judge-facing assessment of RESERVE

Dimension	Evidence-based case
Climate impact	Addresses transport-related emissions in resource routing. Measured real-world impact has not yet been established.
Innovation	Combines feasibility screening, explainable ranking, baseline reconciliation, and evidence governance in one prototype.
Technical execution	Provides a public codebase, live demonstration, deterministic decision logic, local hash-chain mechanism, and Solidity prototype. The scope of working and unverified components is disclosed.
Practical usefulness	Offers a plausible workflow for organizations handling resource shipments, subject to integration with real processor and logistics data.
Presentation	The workflow, plots, tables, source links, assumptions, and validation gaps are documented so reviewers can inspect the reasoning behind the results.

Third, the decision model uses fixed illustrative weights. Its scores are advisory and have not been calibrated against operational outcomes. The implementation demonstrates explainable weighted ranking rather than a trained or independently validated predictive model.

Fourth, the local SHA-256 hash chain should not be described as a decentralized blockchain. The repository's Solidity contract is a prototype; the available evidence does not establish public-chain deployment or live on-chain activity. A production-grade audit system would also require a documented persistence strategy, independent verification, appropriate key management, and a clear record-governance policy.

Finally, the current evidence does not establish measured emissions reductions. The research audit reproduces a synthetic scenario and calculates factor sensitivity, but it does not provide operational shipment measurements or a verified counterfactual. These limitations are not resolved by the presence of a functioning interface or a cryptographic record.

XIV. Path from Prototype to Operational Validation

The next phase should focus on closing the evidence gaps in a deliberate order. More sophisticated optimization or additional visual features would have limited value if the underlying operational data and emissions assumptions remained uncertain.

These steps provide a practical route toward a stronger climate-impact claim. They also preserve the distinction between the current prototype and a future operational system. The application need not be described as fully validated in order to demonstrate why its workflow is useful or what evidence would make it more credible.

XV. Conclusion

RESERVE addresses a practical question in climate-oriented resource management: how can a routing recommendation

TABLE IX
RECOMMENDED VALIDATION ROADMAP

Stage	Work required	Purpose
1	Obtain real processor records	Verify material compatibility, capacity, quality, and availability.
2	Confirm transport activity	Document mode, actual route, distance definition, load, and shared-versus-dedicated freight assumptions.
3	Approve an applicable factor	Match source, units, vehicle category, allocation method, and emissions boundary to the intended use.
4	Define a defensible baseline	Compare equivalent shipments against a feasible, documented counterfactual.
5	Run operational pilots	Record actual shipments, selected routes, delivered quantities, and relevant emissions data.
6	Validate the decision model	Assess recommendations against operational outcomes and document the rationale for weights and evaluation metrics.
7	Strengthen audit infrastructure	Document persistence, verification, access controls, and any actual public-chain integration before making stronger trust claims.

account for feasibility, transport-related emissions, multiple decision criteria, and the credibility of the evidence behind its result?

The project offers a working prototype that screens processor options, estimates transport-related emissions, ranks eligible candidates using a transparent weighted model, and explores cryptographic record traceability. Its public repository makes the implementation open to inspection, while the separate research audit provides a more careful account of the assumptions and limitations behind the demonstration.

The findings are deliberately specific. The synthetic scenario selects Processor B, but the nearest-eligible baseline also selects Processor B, producing zero modeled improvement. The EPA-derived candidate factor changes the absolute estimates but does not change the ranking under a common positive factor. Because the candidate's applicability has not been established, it remains a sensitivity case rather than an adopted factor. No measured real-world emissions reductions have been verified.

For the IEEE ClimateChain Global Hackathon, RESERVE offers a concrete exploration of sustainable supply-chain decision support, explainable AI, and evidence accountability. Its contribution is not that it has already demonstrated climate savings, but that it shows how a routing decision can be made more inspectable and how environmental claims can be subjected to explicit checks.

The next stage is clear: replace synthetic inputs with verified operational records, establish an applicable emissions methodology, test the recommendations against a defensible baseline, and measure actual outcomes. Those steps would determine whether the potential of the current prototype translates into meaningful environmental benefit.

PROJECT LINKS AND REPRODUCIBILITY

The application and the research audit serve complementary purposes. The public repository contains the prototype implementation and related source code. The live website demonstrates the application workflow. The separate research audit examines the evidence available for interpreting the demonstration. The audit should not be confused with a deployed application feature, and its outputs should not be assumed to be displayed on the live website.

- **Live demonstration:** https://reserve-climatechain.onrender.com/
- **Public source repository:** https://github.com/u19725505-star/reserve-climatechain-2026
- **IEEE ClimateChain challenge:** https://ieee-climatechain-hack.devpost.com/
- **EPA 2025 GHG Emission Factors Hub:** https://www.epa.gov/system/files/other-files/2025-01/ghg-emission-factors-hub-2025.xlsx

The numerical results in this report are reproducible calculations from the stated synthetic scenario and assumptions. They should not be interpreted as operational measurements. The report records the saved test summary as previously reported and does not claim an independent rerun of the full test suite.

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